

# RUDRA CHAVDA

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## EDUCATION

**University of California, Santa Barbara**

Expected Spring 2028

*Bachelor of Science, Computer Engineering*

**University of California, Santa Cruz**

2024 – 2026

*Bachelor of Science, Computer Engineering*

**GPA: 3.95**

– **Honors:** 6x Dean's List

**Coursework:** Systems programming, computer architecture, operating systems, data structures and algorithms, and software engineering.

## TECHNICAL SKILLS

**Languages** Python, C, C++, Java, TypeScript, JavaScript, SQL, Assembly

**Frameworks** Vue.js, React, Next.js, Node.js, Spring Boot, Py4Web, Tailwind CSS, Angular

**Tools** Git, Github, MySQL, Flyway, Microsoft Graph API, JUnit, Docker, Nautilus (Cluster), CARLA, GDAL

**Concepts** LLM Agents, Function Calling, OAuth 2.0, Reinforcement Learning, System Design, Data Structures, Algorithms

## EXPERIENCE

**Ascend Cargo**

Jun 2026 – Sep 2026

*AI/ML Intern*

- Built an **LLM agent framework (Java 23, Spring Boot 3.4, OpenAI function calling)** with annotation-driven tool registration, automatic JSON-Schema generation, and LLM-based intent routing that scopes which tools each request may reach.
- Shipped natural-language **create/update for freight shipments and orders**, replacing multi-screen data entry; diagnosed and fixed a synchronous automation call that blocked shipment creation for **12 minutes, cutting it to under 30 seconds**.
- Designed a **two-gate authorization model** enforcing a per-user AI access ceiling (**NONE/READ/WRITE**) alongside existing app privileges, guaranteeing the agent can never exceed the permissions of the human it acts for.
- Delivered an **inbound email channel over the Microsoft Graph API (OAuth 2.0 client credentials)** letting verified staff create and modify shipments by emailing a shared mailbox, with sender verification resolved per-user from the database.
- Hardened agent behavior after live-demo failures: replaced silent fallbacks with **refuse-and-instruct** error contracts that tell the model how to recover, eliminating a class of confidently-wrong records.
- Authored **122 unit tests (JUnit 5, Mockito)** plus a CLI harness that drives the chat API and verifies results against the database, and integrated 3 months of upstream divergence across 38 conflicting files.

**AIEA Lab (UCSC)**

Dec 2025 – Apr 2026

*Undergraduate Researcher*

- Wrote and benchmarked Reinforcement Learning (RL) algorithms such as **DQN, DDQN, A3C**, and **Stable Baselines** in the Car Racing environment to evaluate autonomous vehicle safety as part of the Robustifying AV's project.
- Deploying the **CARLA** simulator in a headless environment on the **Nautilus** hyper-cluster (Kubernetes).
- Analyzed edge-case failures by visualizing agent behaviors and reward functions post-training.

**CA-Futures (UCSC EEB & Baskin Engineering)**

May 2025 – Sep 2025

*Software Developer*

- Developed a full-stack conservation modeling application using **Vue.js** (frontend) and **Py4Web** (backend).
- Integrated geospatial analysis tools (**GDAL**, Ecoscape) to track species connectivity and visualize ecological data.
- Optimized simulation parameters to process large-scale biological datasets for the Ecology Lab.

**Opennote (YC S25)**

Jul 2025 – Feb 2026

*Campus Growth Lead*

- Co-organized UCSC Cruzhacks in collaboration with Opennote where we gave out \$1500 worth of prizes.
- Organized workshops talking about user-agent API flow in typescript.
- Spearheaded campus growth strategy, increasing user engagement by 30% and acquiring 50+ active users.

- Architected a secure, interactive web application using **React.js** and **Tailwind CSS**.
- Designed a responsive front-end framework to improve data visualization for client dashboards.

## PROJECTS

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### Better Notes | *TypeScript, Tailwind CSS, Vercel*

Spring 2025

- Engineered a full-stack note-taking platform with a custom markdown rendering engine for student study workflows.
- Secured 100+ active users within 3 months of deployment to Vercel.
- Live Demo: [betternotes-two.vercel.app](https://betternotes-two.vercel.app)

### Div Explorer Project | *Machine Learning, Python*

Fall 2025

- Developed actionable data labeling pipelines to detect duplicate discoveries and high-overlap patterns.
- Implemented categorization metrics (interpretability, coverage, anomalies) to improve ML model fairness.

### Personal Portfolio | *Next.js, Three.js, Motion*

Winter 2024

- Built a high-performance portfolio site featuring 3D interactive elements using Three.js and Motion animations.

## LEADERSHIP

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### Eagle Scout Project

Summer 2024

*Project Lead, Troop 537*

- Directed a team of 20+ Scouts in the construction of redwood structures for Sycamore Grove Park (Livermore).